Verifiable Voice Protocol

Abstract

Verifiable Voice Protocol (VVP) authenticates and authenticates organizations and individuals making telephone calls. This document specifies the protocol, its cryptographic primitives, and the data structures that carry it on the wire. It defines a hierarchical trust model rooted in a Root Certificate Authority, a delegation chain that lets brands trust different technical and government assurance providers, and case storage. Under evidence, the attestation chain is verifiable offline by any party holding the Root CA public key. Revocation is handled by short-lived certificates, not by revocation lists. Revocation is scoped to a brand. VVP can layer on top of STIR/SHAKEN without changing interconnection agreements or carrier infrastructure. The protocol is designed for deployment by telephony providers, brands, and verification services, with a clear path to interoperability with existing caller ID authentication systems.

About This Document

This document is not to be distributed anywhere else.

The latest revision of this draft can be found at https://drafts.ietf.org/drafts/draft-hallambaker-verifiable-voice-protocol/draft-hallambaker-verifiable-voice-protocol.txt

Discussion of this document takes place on the Secure Telephone Identity Revisited (stir) mailing list.

Source for this draft and an issue tracker can be found at https://github.com/dkg/stir-drafts/.

Status of This Memo

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1. Introduction

Voice calls are not authenticated. Caller ID is trivially spoofed, and there is no cryptographic binding between the call and the entity that placed it. Telephony fraud, robocalls, and impersonation calls cost billions annually. Existing mitigations such as STIR/SHAKEN [RFC8224] authenticate the carrier that handles the call, but they do not authenticate the organization that actually placed it, and they do not provide a mechanism for a third party to verify that a call was authorized by a specific brand.

Verifiable Voice Protocol (VVP) addresses this by binding a cryptographic attestation to the call, issued by a signature authority that the terminating party can verify offline against a root certificate it holds. The attestation carries the identity of the originating service provider, with no independent assurance of the underlying caller's identity. Stronger assurance is provided by delegation chains that let brands trust specialized assurance providers without bilateral agreements with each carrier.

VVP can layer on top of STIR/SHAKEN without changing interconnection agreements or carrier infrastructure. The protocol is designed for deployment by telephony providers, brands, and verification services, with a clear path to interoperability with existing caller ID authentication systems.

1.1. Evidence Scope

The protocol proves that a specific signaling service attested to a caller's identity, plus provably correct binding of attributes. These assertions are verified by a carrier-provisioned Logotype and are meaningful only in the context of that carrier's brand relationship with its subscribers.

VVP does not claim to independently verify caller identity. The attestation authority attests what it knows about the caller's identity, and the strength of that attestation depends on the underlying assurance process. Brands that need stronger assurance can require higher assurance levels in their delegation chains, but the protocol itself does not perform identity proofing.

1.2. Relationship to STIR/SHAKEN

VVP is designed to coexist with STIR/SHAKEN. Carriers that deploy STIR/SHAKEN can add VVP attestation without changing their interconnection agreements. The Signaling Protocol (Section 3.5) carries VVP tokens in parallel with SIP headers, so intermediaries that do not understand VVP can ignore the extension and pass the call through unchanged.

2. Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC8174] when, and only when, they appear in all capitals, as shown here.

3. Overview

Fundamentally, VVP creates a chain in order to assert (Section 3.1.1) that a signaling service (the phone) is authorized to place calls for a specific identity. The chain has five levels: a root certificate, an attestation authority certificate, a terminating party certificate, a signature authority certificate, and an attestation token. The root certificate is held by the terminating party, the attestation token is presented with the signaling message, revocation tokens let parties check for revocation, and the signature authority certificate is the link between the root and the attestation token.

3.1. Hashes

Cryptographic hashes are used in several places in VVP. The hash algorithm is identified by a COSE hash algorithm identifier [RFC9053], currently SHA-256 (-16) and SHA-512 (-17). Implementations MUST support SHA-256 and SHA-512.

3.2. Signatures

The signature algorithm is identified by a COSE algorithm identifier [RFC9053], currently ES256 (-7) and EdDSA (-8). Implementations MUST support ES256 and EdDSA.

3.3. Attestation Identifier

The attestation identifier carries the phone number or URI being attested. It is a tagged union: one variant for a phone number in E.164 format, one for a URI. The identifier is the subject of the attestation token and the target of revocation.

3.4. Terminating Party

The terminating party is the verifier. It holds the root certificate and checks the entire chain. The protocol assumes the terminating party can obtain revocation lists for the authorities it trusts.

3.5. Signaling Protocol

The signaling protocol carries VVP tokens between the originating service provider and the terminating party. The protocol is out of scope for this document. The Signaling Protocol section (Section 5) defines how tokens are presented and received.

3.6. Attestation Tokens

The attestation token is the artifact presented with a call. It carries the attestation identifier, the identity of the originating service provider, and optional fingerprints of the call's parties. It is signed by the signature authority and verifiable against the root certificate.

3.7. Revocation Tokens

Revocation tokens let parties check whether a certificate has been revoked. They are short-lived assertions that a specific certificate was revoked at a specific time. Revocation is scoped to a brand: a revocation from one brand does not affect certificates issued by other brands.

3.8. Case Bundle

A case bundle packages all the evidence for a call into a single file: the chain, the attestation token, and revocation tokens. This simplifies transport and storage. The bundle is self-contained: a verifier needs only the root certificate and the bundle to validate the call.

4. Chain

A valid attestation chain consists of the root certificate, a brand certificate, a signature authority certificate, and a terminating party certificate. The root certificate is held by the terminating party. The other three certificates are presented with the call. The chain proves that the root authorized a brand to delegate to a signature authority, that the signature authority was authorized to issue attestations, and that the terminating party's certificate is valid for the subject of the attestation.

4.1. Root Certificate

The root certificate is a self-signed certificate that anchors the chain. It MUST have the keyCertSign and cRLSign key usages. The root MAY carry a brand list in a private extension; if it does, the brand list MUST be a subset of the parent's brand list.

4.2. Attestation Authority Certificate

The attestation authority certificate is issued by the root to a brand. It MUST have the keyCertSign and cRLSign key usages. The brand list MUST be a subset of the parent's brand list.

4.3. Terminating Party

The terminating party certificate is issued by the root. It MUST have the keyCertSign and cRLSign key usages. The brand list MUST be a subset of the parent's brand list.

4.4. Signature Authority

The signature authority certificate is issued by the terminating party. It MUST have the digitalSignature key usage. The brand list MUST be a subset of the parent's brand list.

4.5. Revocation Authority

The revocation authority certificate is issued by the root. It MUST have the cRLSign key usage. The brand list MUST be a subset of the parent's brand list.

5. Attestation

The attestation token is presented with a call. It carries the attestation identifier, the identity of the originating service provider, and optional fingerprints of the call's parties. It is signed by the signature authority and verifiable against the root certificate.

5.1. Signature Authority

The signature authority signs the attestation token. It MUST be issued by a certificate in the chain with the digitalSignature key usage. The token's issuer field MUST match the subject of the signature authority certificate.

5.2. Attestation Authority

The attestation authority issues the signature authority certificate. The brand list in the signature authority certificate MUST be a subset of the attestation authority's brand list.

6. Revocation

Revocation tokens let parties check whether a certificate has been revoked. They are short-lived assertions that a specific certificate was revoked at a specific time. Revocation is scoped to a brand: a revocation from one brand does not affect certificates issued by other brands.

6.1. Revocation Authority

The revocation authority issues revocation tokens. It MUST be issued by a certificate in the chain with the cRLSign key usage. The token's issuer field MUST match the subject of the revocation authority certificate.

7. Verifying

7.1. Algorithm

When a call arrives, the terminating party:

1. Extracts the attestation token from the signaling protocol.
2. Extracts the attestation identifier, the issuer, and the issuance time from the token.
3. Selects the trusted root.
4. Verifies the chain: checks each certificate's signature, validity period, and key usage against its parent, and checks that the issuer field of each certificate matches the subject of its parent.
5. Checks revocation: looks up each certificate in the chain by issuer and serial number in the revocation list, and rejects if any match.
6. Verifies the token: checks the token's signature against the signature authority certificate, and checks that the issuance time falls within the signature authority certificate's validity period.
7. Matches the offering: checks that the attestation identifier matches the phone number or URI being called.

7.2. Matching the Offering

The complete algorithm is more involved than the steps above. In particular, the matching step supports wildcards and multi-party calls. This section will be expanded in a future revision.

8. Case Bundle

A case bundle packages all the evidence for a call into a single file: the chain, the attestation token, and revocation tokens. This simplifies transport and storage. The bundle is self-contained: a verifier needs only the root certificate and the bundle to validate the call.

8.1. Algorithm

1. Collects the chain: gathers the brand certificate, the signature authority certificate, and the attestation authority certificate.
2. Collects the attestation token.
3. Collects revocation tokens: looks up each certificate in the chain by issuer and serial number in the revocation list.
4. Packs into a CBOR file: serializes the chain, token, and revocation tokens into a single CBOR file.

8.2. Matching the Offering

The complete algorithm is more involved than the steps above. In particular, the matching step supports wildcards and multi-party calls. This section will be expanded in a future revision.

6.2.7.5. D

The Delegated Credential is a tentative, frame of time, to a sender's DC that qualifies what the TLS 1.3 processing, to use its Delegated Credential, or not. Its value is an UINT32 in units of 10 seconds.

6.3. Specific artifacts

6.3.1. PSL

Each root cert begins with a DER SEQUENCE, and a DER-encoded UTF8 Currency or Date is a tagged, fixed-length, 10-byte, 10-digit, 10-digit, 10-digit

12.6. Delegated signer credential

A delegated signer credential (6.3.9) must prove that the issuer is giving authority to the issuee. This authority should be carefully constrained so that it applies only to outbound voice calls, not to signing invoices or legal contracts. It can also be constrained so it only applies on a particular schedule, or when the call originates or terminates in a particular geo or jurisdiction.

A Generalized Cooperative Delegation (GCD) credential embodies delegated but constrained authority in exactly this way. A GCD credential suitable for use as a delegated signer credential in VVP might look like this:

```
{
    "v": "ACDC10JSON00096c_",
    "d": "EDQpUInrKyJBgUJGw5461CbWcug9BZj7WXUkKbNO1nFR",
    "u": "0ADE6oAxB14E7uKeOUb7BEYi",
    "i": "ECA5uEyzrRwu3FBFrKR0bd9xejloSbUwAtGcbBGUk2nL",
    "rl": "EM2YZ78SKE8eO4M1QOJeerSxKZqLmJV7SPr3JiSDMBZ",
    "s": "EL7irIKYJL9Io0hhKSGWI4Oznhwc7qgJG5Qf4aEs6j0o",
    "a": {
        "d": "EPQhEk5tfXvvyKeSHk4DG63dSgoP-F2VZrxuIeIKrT9B",
        "u": "0ACRkH8q99PTCQNte0yI-f4g",
        "i": "EIkxoE8eYnPLCydPcyc_lhQgwOdBHwzkSe36e2qqEH-5",
        "dt": "2024-12-27T13:11:29.062000+00:00",
        "c_goal": ["ops.telco.send.sign"],
        "c_proto": ["VVP:OP"]
    },
    "r": {
        "d": "EFthNcTE20MLMaCOoXlSmNtdooGEbZF8uGmO5G85eMSF",
        "noRoleSemanticsWithoutGfw": "All parties agree...",
        "issuerNotResponsibleOutsideConstraints": "Although verifiers...",
        "noConstraintSansPrefix": "Issuers agree...",
        "useStdIfPossible": "Issuers agree...",
        "onlyDelegateHeldAuthority": "Issuers agree..."
    }
}
```

Note the `c_goal` field that limits goals that can be justified with this credential, and the `c_proto` field that says the delegate can only exercise this authority in the context of the "VVP" protocol with the "OP" role.

The schema for GCD credentials, and an explanation how to add additional constraints, is documented at [GCD-SCHEMA].

12.7. Dossier

A dossier (6.3.2) is almost all edges -- that is, links to other credentials. Here's what one might look like:

```
{
  "v": "ACDC10JSON00036f_",
  "d": "EKvpcshjgjzdCWwR4q9VnlsUwPgfWzmy9ojMpTSzNcEr",
  "i": "EIkxoE8eYnPLCydPcyc_lhQgwOdBHwzkSe36e2qqEH-5",
  "ri": "EMU5wN33VsrJK1GCwk2ts_LJi67IXE6vrYV3v9XdxwSp",
  "s": "EFv3_L64_xNhOGQkaAHQTI-lzQYDv1aHcuZbuOTuDBXj",
  "a": {
    "d": "EJnMnz8MJxmI0epkq7DtzzPSgtDbSD2YxkSdczNFcHQM",
    "dt": "2024-12-27T13:11:41.865000+00:00"
  },
  "e": {
    "d": "EPVc2ktYnZQOwNs341OYXFOkTS1IzaILFEMXNfZbGrh",
    "vetting": {
      "n": "EIpaOx1NJc0N_OjSxzWHfQp6EpB847yTex62xQ7uuSQL",
      "s": "ENPXp1vQzRF6JwIuS-mp2U8Uf1MoADoP_GqQ62VsDZWY",
      "o": "NI2I"
    },
    "alloc": {
      "n": "EEeg55Yr01gDyCScFUaE2QgzC7IOjQRpX2sTckFZp1RP",
      "s": "EFvnoHDY7I-kaBBeKlbDbkjG4BaI0nKLGadxBdjMGgSQ",
      "o": "I2I"
    },
    "brand": {
      "n": "EKSZT4yTtaZ2Aqr1NKBvS7GjmsU3Xlt-S3c69pHceIXW",
      "s": "EaRmoHDYbkjG4BaI0nK7I-kaBBeKlbDLGadxBdSQjMGg",
      "o": "I2I"
    },
    "bproxy": {
      "n": "EMXpU3nrKyJBgUJGw5461CbWcug9BZj7WXUkKbNO1nFx",
      "s": "EL7irIKYJL9Io0hhKSGWI4Oznhwc7qgJG5Qf4aEs6j0o",
      "o": "I2I"
    },
    "delsig": {
      "n": "EMKcp1-AvpW0PZdThjK3JCbMsXAmrqB9ONa1vZvTpgOE",
      "s": "EL7irIKYJL9Io0hhKSGWI4Oznhwc7qgJG5Qf4aEs6j0o",
      "o": "NI2I"
    }
  }
}
```

Notice how each named edge references one of the previous sample credentials in its n field, and that other credential's associated schema in the s field.

The schema for this credential is documented at [DOSSIER-SCHEMA].

13. IANA Considerations

This specification requires no IANA actions. However, it does depend on OOBIs (see 6.2.3) being served as web resources with IANA content type `application/json+cesr`.

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